



MONOPOLY in LK

Take Home Assignment

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1 The MONOPOLY-LK Board

MONOPOLY-LK is a strategic property trading and economic simulation game inspired by the traditional game of Monopoly. The objective of the game is to become the wealthiest player through purchasing, developing, leasing, mortgaging, insuring, and managing real estate while avoiding bankruptcy. Unlike the traditional game, MONOPOLY-LK models several aspects of the Sri Lankan economy, introducing banking, insurance, taxation, and loan management into the gameplay.

The game is played entirely by four autonomous computer controlled players, each possessing a unique investment strategy and financial behaviour. Once the simulation begins, no user interaction is required. The players independently make all decisions regarding purchasing properties, obtaining loans, purchasing insurance, constructing buildings, participating in auctions, and managing their financial resources.

The game consists of a square board containing forty (40) spaces. During each turn, players roll two six-sided dice and move around the board in a clockwise direction. Depending on the square landed upon, a player may purchase property, pay rent, receive income, obtain a loan, purchase insurance, pay taxes, participate in auctions, draw event cards, or be sent to jail.

Unlike the traditional Monopoly game, MONOPOLY-LK incorporates realistic financial systems commonly found in Sri Lanka. Players may obtain loans from commercial banks by mortgaging properties as collateral. Loans accumulate interest every round and may eventually lead to foreclosure if repayments are not made. Insurance companies offer various insurance policies that protect players from unexpected disasters including fires, floods,

riots, and property damage. Throughout the game, random economic events may affect the national economy, causing fluctuations in property values, interest rates, construction costs, and rental income.

The winner is the player with the highest net worth after all opponents become bankrupt or after the maximum number of rounds has been completed. Net worth is calculated by considering available cash, owned properties, developed buildings, insurance compensation receivable, outstanding loans, and mortgage liabilities.

The primary objective of this assignment is to evaluate the student's ability to design efficient data structures, implement complex game logic, simulate autonomous decision making, manage multiple interacting objects, and produce modular, maintainable software using the C programming language.

1.1 The Board

The MONOPOLY-LK board consists of forty (40) squares arranged around the perimeter of a square playing board. The board is divided into various categories including properties, railway stations, utility companies, commercial banks, insurance companies, taxation squares, event squares, jail, and free parking. Every square on the board shall be assigned an index beginning from 0 and continuing clockwise until 39. The layout is shown in Table 1.

1.1.1 Property Groups

Properties belong to one of eight investment groups. A player owning every property within a group owns a monopoly for that group and becomes eligible to develop buildings.

- **Group 1 – Brown**

- Pettah
- Maradana

- **Group 2 – Light Blue**

- Bambalapitiya
- Wellawatte
- Mount Lavinia

- **Group 3 – Pink**

- Maharagama
- Nugegoda
- Kottawa

- **Group 4 – Orange**

- Negombo
- Katunayake
- Ja-Ela

Table 1: Board Layout of Monopoly-LK

Square	Type	Name
0	Start	GO
1	Property	Pettah
2	Event	Community Development Fund
3	Property	Maradana
4	Tax	Income Tax
5	Railway	Colombo Fort Railway Station
6	Property	Bambalapitiya
7	Event	National Event Card
8	Property	Wellawatte
9	Property	Mount Lavinia
10	Special	Jail / Just Visiting
11	Property	Nugegoda
12	Utility	Ceylon Electricity Board
13	Property	Maharagama
14	Property	Kottawa
15	Railway	Kandy Railway Station
16	Property	Negombo
17	Insurance	Sri Lanka Insurance
18	Property	Katunayake
19	Property	Ja-Ela
20	Special	Free Parking
21	Property	Kandy City
22	Event	National Event Card
23	Property	Peradeniya
24	Property	Katugastota
25	Railway	Galle Railway Station
26	Property	Galle Fort
27	Property	Unawatuna
28	Utility	National Water Supply and Drainage Board
29	Property	Hikkaduwa
30	Special	Go To Jail
31	Property	Jaffna Town
32	Property	Nallur
33	Insurance	Ceylinco Insurance
34	Property	Trincomalee
35	Railway	Jaffna Railway Station
36	Event	National Event Card
37	Property	Nuwara Eliya
38	Bank	Bank of Ceylon
39	Property	Galle Face

- Group 5 – Red

- Kandy City

- Peradeniya
- Katugastota

- **Group 6 – Yellow**

- Galle Fort
- Unawatuna
- Hikkaduwa

- **Group 7 – Green**

- Jaffna Town
- Nallur
- Trincomalee

- **Group 8 – Dark Blue**

- Nuwara Eliya
- Galle Face

Each property has the following properties, and initially all of them belong to the bank.

- Purchase price
- Mortgage value
- Base rental
- House construction cost
- Hotel construction cost
- Current owner
- Mortgage status
- Insurance status
- Number of building

1.1.2 Railway Stations

The board contains four railway stations which cannot be further developed. In addition, stations can be mortgaged to the bank if required however, they cannot be insured. Rental for a railway station is dependent on how many railway stations are owned by a single player as per the Table 2 below.

- Colombo Fort Railway Station
- Kandy Railway Station
- Jaffna Railway Station
- Galle Railway Station

Table 2: Railway Stations Rent

Stations Owned	Rent
1	LKR 250
2	LKR 500
3	LKR 1000
4	LKR 2000

1.1.3 Utility Companies

The board contains two public utility companies. Utilities may be mortgaged but cannot be developed.

- Ceylon Electricity Board (CEB)
- National Water Supply and Drainage Board (NWSDB)

Utility companies produce rental based upon the dice value rolled by the visiting player. If the owner possesses :

- one utility, rent equals $4 \times$ dice value
- both utilities, rent equals $10 \times$ dice value

1.1.4 Commercial Banks

Unlike traditional Monopoly, commercial banks exist as board locations. The board contains one commercial bank.

- Bank of Ceylon

Whenever a player lands on the Bank square, the player may perform one financial transaction. The player may:

- obtain a secured loan
- repay an existing loan
- refinance an existing loan
- increase loan amount
- repay loan in full

A player may possess only one active loan at any time. The maximum loan amount equals 75% of the total mortgage value of all unmortgaged properties owned by the player. Loans require collateral. Collateral remains owned by the player but cannot be sold until the loan has been settled. Interest accumulates after every complete round. Failure to repay the loan before its maturity causes foreclosure. During foreclosure:

- mortgaged properties transfer to the Bank
- outstanding debt is cancelled
- buildings are demolished
- insurance policies are cancelled

1.2 Insurance Companies

Two insurance companies operate on the board.

- Sri Lanka Insurance
- Ceylinco Insurance

Landing on an insurance square allows a player to purchase or renew insurance. Insurance policies remain valid for twenty rounds. Three insurance policies exist.

1. Basic Property Insurance

- Protect Against
 - Fire
 - Flood
- Premium
 - 5% of property value.
- Compensation
 - 80% of repair cost.

2. Comprehensive Insurance

- Protect Against
 - Fire
 - Flood
 - Riot
 - Vandalism
- Premium
 - 10% of property value.
- Compensation
 - 100% of repair cost.

3. Business Interruption Insurance

- Compensation - *Applicable only to properties containing hotels*
 - Repair cost
 - Lost rental income for five rounds
- Premium
 - 15% of property value.

Players may insure individual properties independently. Insurance automatically expires after twenty rounds. Players receive notification three rounds before expiry. If a disaster occurs on an uninsured property, the owner bears the full repair cost.

1.3 Traditional Rules

The following rules define the traditional gameplay mechanics used in MONOPOLY-LK. Unless explicitly modified in the later sections describing the MONOPOLY-LK extensions, these rules shall remain in effect throughout the simulation.

Rule 1 : The game is played by four players. Each player begins with

- LKR 30,000 cash
- no properties
- no loans
- no insurance policies
- no railway stations
- no utilities.

Rule 2 : The order of play shall be determined before the game begins. Each player rolls two dice. The player obtaining the highest total begins first. In case of a tie, tied players reroll until the order is determined. Play proceeds clockwise thereafter.

Rule 3 : A player's turn consists of the following sequence:

1. Resolve outstanding penalties.
2. Roll two dice.
3. Move clockwise.
4. Resolve landing action.
5. Purchase property if eligible.
6. Construct buildings if eligible.
7. Complete financial transactions.
8. End turn.

Rule 4 : Passing or landing on GO awards the player LKR 2,000.

Rule 5 : If an unowned purchasable square is landed upon, the player may purchase it at its listed price. If declined, the property immediately enters auction.

Rule 6 : Auctions involve every player except bankrupt players. Bidding increases in increments of LKR 250. The highest bidder acquires ownership.

Rule 7 : Landing on an owned property requires payment of rent to the owner. If the property is mortgaged, no rent shall be collected.

Rule 8 : Owning every property within a colour group establishes a monopoly. Only monopolies permit construction.

Rule 9 : Buildings shall be constructed evenly across properties within the same colour group. No property may exceed:

- four houses
- one hotel

Rule 10 : Hotels replace four houses. A property cannot simultaneously contain houses and a hotel.

Rule 11 : Landing on Income Tax requires payment of the specified tax immediately. Failure to pay follows the normal debt recovery process.

Rule 12 : Landing on Go To Jail immediately transfers the player to Jail. The player does not collect GO money.

Rule 13 : A jailed player may leave by

- paying bail of LKR 300,
- rolling doubles,
- remaining imprisoned for three turns.

Rule 14 : Players may become bankrupt whenever liabilities exceed available assets. Upon bankruptcy

- all buildings are removed,
- insurance policies expire,
- loans become immediately due,
- remaining assets transfer according to the bankruptcy rules defined later in this specification.

Rule 15 : The game concludes when only one player remains solvent or when the maximum simulation limit of 500 rounds has been reached. If the maximum number of rounds is reached, the player possessing the highest net worth is declared the winner.

Net worth is calculated as:

- **Cash + Property Value + Building Value + Railway Value + Utility Value + Insurance Claims Receivable - Outstanding Loans - Accrued Interest - Taxes Due.**

These rules establish the baseline Monopoly mechanics. Subsequent sections of the assignment will introduce MONOPOLY-LK-specific extensions, including dynamic economic events, advanced banking, insurance processing, player strategies, autonomous decision making, and reporting requirements, which supersede or augment these traditional rules where applicable.

2 MONOPOLY-LK Extensions

The following rules extend the traditional Monopoly rules described in Section [1.3](#). Unless explicitly stated otherwise, these rules shall be applied together with the traditional rules throughout the simulation.

Unlike traditional Monopoly, MONOPOLY-LK introduces banking, insurance, economic fluctuations, government intervention, and dynamic property valuation. These extensions are intended to simulate a realistic economic environment while increasing the strategic complexity of the game.

2.1 Loans

Unlike the traditional game, players may obtain secured loans from commercial banks. Loans are only available when a player lands on the Bank of Ceylon square. Each player may maintain only one active loan at any given time.

Rule-LK 1 : A player may borrow money only if sufficient collateral exists. Buildings are not considered collateral. Eligible collateral includes:

- Properties
- Railway Stations
- Utility Companies

Rule-LK 2 : The maximum loan amount equals 75% of the total mortgage value of all eligible collateral. For example,

- If the mortgage values are
 - Property A : LKR 2,000
 - Property B : LKR 3,000
 - Railway : LKR 4,000
- Maximum Loan is
 - $= (2000 + 3000 + 4000) \times 75\% = \text{LKR } 6,750$

Rule-LK 3: A loan immediately credits the approved amount to the player's cash balance. The pledged properties become Loan Locked. Loan Locked properties

- cannot be sold,
- cannot be traded,
- cannot be auctioned,
- cannot be additionally mortgaged,
- continue earning rent,
- may still be developed.

Rule-LK 4: Loans have an initial duration of 20 rounds. At the end of every complete round, interest is accumulated. Interest is compounded according to $\text{Current Loan} \times \text{Current Interest Rate}$. The accumulated interest becomes part of the outstanding loan.

Rule-LK 5: A player landing on the Bank may perform one of the following actions.

- Repay part of the loan
- Repay the entire loan
- Extend the loan period
- Increase the loan amount (subject to collateral availability)

Rule-LK 6: Failure to repay a loan within its duration causes Loan Default. When default occurs

- all pledged assets immediately transfer to the Bank,
- hotels and houses are demolished,
- insurance policies on those properties are cancelled,
- outstanding debt is cleared,
- the player continues the game using any remaining assets.

Rule-LK 7: If all pledged assets are foreclosed and the player possesses no remaining assets, the player is declared bankrupt.

2.2 Insurance

Insurance protects players from disasters and unexpected economic losses. Insurance can only be purchased by landing on either

- Sri Lanka Insurance
- Ceylinco Insurance

Each policy applies only to one property.

Rule-LK 8: Three insurance policies exist as specified in Section 1.2

Rule-LK 9: Insurance expires after twenty rounds. Players receive renewal reminders three rounds before expiry. Expired policies provide no protection

Rule-LK 10: Random disasters may occur every ten rounds. Possible disasters include

- Fire
- Flood
- Riot
- Building Collapse
- Electrical Failure

Disasters affect one randomly selected developed property. If insured, compensation is immediately credited. Otherwise, repair costs must be paid by the owner.

Rule-LK 11: Damaged buildings cannot collect rent until repairs are completed. Repairs are automatically performed when the owner has sufficient funds.

2.3 Inflation

Unlike traditional Monopoly, the Sri Lankan economy changes over time. Inflation affects nearly every financial aspect of the game. Inflation has a compounding effect.

Rule-LK 12: Every ten rounds an inflation rate is generated. Possible values are (Negative values indicate deflation)

- -3%
- 0%

- 2%
- 5%
- 8%
- 12%

Rule-LK 13: Inflation modifies

- Property prices
- Building costs
- Hotel costs
- Rental values
- Insurance premiums
- Repair costs
- Loan interest rates. Existing loan rates remain unchanged.

Rule-LK 14: All values that are effected by inflation are recalculated using **New Value = Previous Value $\times$ (1 + Inflation Rate)**

2.4 Property Depreciation

Buildings naturally lose value as time passes.

Rule-LK 15: Every property records its age. Property age increases every complete round.

Rule-LK 16: Properties older than fifty rounds without renovation lose value with a depreciation of 1% every five rounds up until a maximum depreciation of 30%.

Rule-LK 17: Landing on an owned property allows the owner to renovate. Renovation restores depreciation, increases rental, resets property age. Renovation cost equals 10% of the property's current market value.

2.5 Economic Events

Sri Lanka's economy experiences unpredictable changes. Every fifteen rounds one national event occurs. Events affect every player.

Possible economic events include

Rule-LK 18 : Possible economic events include

- Tourism Boom
 - Hotels receive double rent.
 - Southern coastal properties increase by 15%.
- Fuel Crisis
 - Railway rent doubles.
 - Property development costs increase 20%.
- Heavy Monsoon
 - Flood risk increases.

- Insurance premiums increase.
- Coastal properties lose 10% value.
- Economic Recession
 - Property values decrease 15%.
 - Rent decreases 10%.
 - Loan interest increases by 15%.
- Stock Market Boom
 - Property values increase 10%.
 - Loan interest decreases by 10%.
- Government Housing Programme
 - House construction costs reduce 25%.
- Foreign Investment
 - Commercial properties increase 20%.
- Political Unrest
 - Riot probability doubles.
 - Hotel occupancy decreases and the hotel rent drops by 50%.
 - Business interruption claims increases.

2.6 Auctions

Auctions occur whenever

- a property is declined,
- a bankrupt player's assets are liquidated,
- foreclosed assets return to the Bank.

Rule-LK 19 : All solvent players participate. Bidding begins at 50% of market value.

Rule-LK 20 : Minimum bid increment LKR 250

Rule-LK 21 : A player withdraws permanently after declining to bid. Highest remaining bidder wins.

Rule-LK 22 : A player cannot bid more cash than currently available. Loans cannot be obtained during auctions.

Rule-LK 23 : If no player bids, ownership remains with the Bank.

2.7 Government Regulations

The Sri Lankan Government periodically introduces regulations affecting the economy. Every twenty rounds one regulation is selected randomly.

Rule-LK 24 : Government regulations include:

- Increase Property Tax

- Income Tax increases by 50%.
- Reduce Loan Interest
 - Interest decreases by 2%.
- Housing Subsidy
 - House construction costs reduce 30%.
- Luxury Property Tax
 - Hotels incur an annual maintenance tax of 25% of property value with developments.
- Railway Modernization
 - Railway rents increase 25%.
- Electricity Tariff Revision
 - Utility rents increase 20%.
- Insurance Regulation
 - Insurance premiums decrease 15%.
 - Coverage remains unchanged.
- Anti-Speculation Act
 - Players may own at most three undeveloped properties.
 - Additional purchases require immediate development within five rounds.

2.8 Building Depreciation and Maintenance

Unlike traditional Monopoly, buildings deteriorate with age. Every constructed house and hotel requires periodic maintenance to preserve its value and rental income.

Rule-LK 25 : Every house and hotel shall maintain a Condition Rating. Each building begins with a condition of 100%. The condition decreases by 2% at the end of every round.

Rule-LK 26 : Building condition directly affects rental income as specified in the Table 3. A closed building cannot collect rent until maintenance has been completed.

Table 3: Building Condition Table

Condition	Rent Collected
90–100%	100%
75–89%	90%
50–74%	75%
25–49%	50%
Below 25%	Building Closed

Rule-LK 27 : Players may perform maintenance only at the beginning of their turn. Maintenance restores a building's condition to 100%. Maintenance Costs are as follows:

- House = 5% of construction cost
- Hotel = 8% of construction cost

The player may maintain any number of buildings provided sufficient funds are available.

Rule-LK 28 : If maintenance is ignored for more than 20 consecutive rounds, structural damage occurs. Structural damage immediately

- reduces property value by 15%,
- reduces maximum rent by 25%,
- increases future maintenance costs by 50%.

Rule-LK 29 : A damaged building may be renovated. Renovation restores the following but costs 25% of the hotel's or building's replacement value.

- property value,
- rental value,
- building condition,

2.9 Dynamic Property Market

Unlike traditional Monopoly, property prices are not fixed throughout the game. Different regions of Sri Lanka experience property booms and market declines based on changing economic conditions.

Rule-LK 30 : Every 10 rounds, the Property Market is reviewed. A random property group is selected for a Market Boom. Another property group is selected for a Market Decline. The same property group cannot experience the same event in consecutive reviews.

Rule-LK 31 : During a Market Boom (The boom remains active for 10 rounds.)

- Purchase prices increase by 15%
- Mortgage values increase by 15%
- Rental income increases by 25%
- Construction costs increase by 10%
- Property values increase by 20%

Rule-LK 32 : During a Market Decline (The decline remains active for 10 rounds.)

- Property values decrease by 15%
- Rental income decreases by 20%
- Mortgage values decrease by 10%
- Auction starting prices decrease by 25%

Rule-LK 33 : A property group affected by a boom or decline cannot be selected again until at least 30 rounds have elapsed.

Rule-LK 34 : If multiple economic events affect the same property group simultaneously, all percentage changes are cumulative.

2.10 Regional Development Cards

Every fifteen (15) rounds, a Regional Development Card is drawn. Unlike National Event Cards, these affect only specific regions of Sri Lanka. One Regional Development Card remains active for 15 rounds.

Table 4: Regional Development Cards

Card	Effect
Southern Tourism Boom	Galle Fort, Unawatuna and Hikkaduwa rental income +40%
Port City Expansion	Pettah, Maradana and Colombo Fort Station values +25%
IT Industry Growth	Maharagama, Nugegoda and Kottawa values +20%
Northern Development Programme	Jaffna Town, Nallur and Trincomalee values +30%
Tea Export Boom	Nuwara Eliya value +35%
Airport Expansion	Negombo, Katunayake and Ja-Ela rents +30%
University City Growth	Peradeniya and Kandy City values +20%
Beach Pollution	Southern coastal rents -30%
Flood Damage	Low-lying coastal properties lose 20% value
Transport Strike	Railway revenue reduced by 40%
Electricity Tariff Increase	Utility rent +25%
Water Shortage	Water utility revenue +20%; surrounding properties -10%

Rule-LK 35 : At the expiration of the Regional Development period, all affected property values return to their normal market-adjusted values unless another active event is still influencing them.

Rule-LK 36 : The simulation shall display the currently active regional market conditions at the end of every round.

3 Player Behaviours

The MONOPOLY-LK simulation consists of four autonomous players. Once the simulation begins, no user interaction is permitted. Each player independently makes decisions based on its assigned strategy. These strategies influence every financial and gameplay decision, including property purchases, bidding in auctions, loan management, insurance purchases, construction of buildings, renovations, and asset liquidation. Whenever multiple actions are available during a turn, the player shall evaluate all legal actions and execute the one that best satisfies its behavioural strategy.

3.1 Aggressive Investor

The Aggressive Investor seeks to maximize long-term wealth through rapid expansion and property development. This player prioritizes acquiring monopolies and generating rental

income rather than maintaining large cash reserves. The behaviour of this player is as follows.

- Always purchases an unowned property if sufficient funds remain to pay at least one future rent.
- Always participates in auctions.
- Bids aggressively until the property reaches 120% of its estimated market value.
- Prioritizes completing property groups over acquiring isolated properties.
- Constructs the maximum possible number of houses immediately after obtaining a monopoly.
- Converts houses into hotels as soon as legally permitted.
- Obtains loans whenever additional funds can increase projected rental income.
- Repays loans only when excess cash exceeds twice the outstanding loan amount.
- Purchases only Basic Insurance for houses and Comprehensive Insurance for hotels.
- Never voluntarily sells a property unless bankruptcy is unavoidable.
- Prioritizes acquiring premium properties such as Galle Face and Nuwara Eliya.

3.2 Conservative Banker

The Conservative Banker prioritizes financial stability and minimizes unnecessary risks. Maintaining positive cash flow is considered more important than acquiring numerous properties. The behaviour of this player is as follows.

- Purchases properties only if at least 50% of current cash remains after purchase.
- Participates in auctions only when bidding below market value.
- Avoids obtaining loans unless bankruptcy is imminent.
- Repays loans immediately whenever visiting the Bank if sufficient funds exist.
- Always purchases Comprehensive Insurance for every developed property.
- Never develops hotels until all outstanding loans have been settled.
- Prefers railway stations and utility companies due to their predictable income.
- Avoids investments during economic recessions.
- Renovates depreciated properties immediately once depreciation exceeds 10%.
- Maintains the largest emergency cash reserve among all players.

3.3 Risk Taker

The Risk Taker attempts to maximize returns through highly speculative investments. The player is willing to incur significant debt in pursuit of rapid expansion. The behaviour of this player is as follows.

- Purchases every available property whenever legally possible.
- Always borrows the maximum loan permitted.
- Frequently refinances loans to increase available capital.
- Participates aggressively in every auction.
- Bids until available cash is exhausted.
- Constructs hotels as early as possible.
- Purchases insurance only after experiencing a financial loss.
- Ignores property depreciation until repair becomes unavoidable.
- Sells lower-value properties to finance premium developments.
- Continues investing even during economic downturns.
- Prioritizes expensive property groups over balanced portfolios.

3.4 Opportunistic Trader

The Opportunistic Trader continuously evaluates market conditions before making investment decisions. Unlike the other players, this strategy adapts dynamically to economic events and government regulations. The behaviour of this player is as follows.

- Purchases properties only when projected appreciation exceeds construction costs.
- Prefers discounted auction purchases rather than direct purchases.
- Obtains loans only when projected return exceeds borrowing cost.
- Purchases Comprehensive Insurance only for high-value developments.
- Delays construction during inflation.
- Accelerates construction during Government Housing Subsidy periods.
- Renovates properties once depreciation exceeds 15%.
- Sells properties expected to decrease in value following economic events.
- Prioritizes investments benefiting from current government regulations.
- Maintains a balanced portfolio consisting of residential properties, railway stations, and utilities.
- Always evaluates expected return before making any financial decision.

4 Program Requirements

Students are required to implement the complete MONOPOLY-LK simulation using the C Programming Language. The simulation shall execute entirely through the commandline interface without requiring any user interaction once execution has begun. The implementation shall conform to all rules specified in Sections 1 through 3. The following requirements shall be satisfied.

- The program shall simulate four autonomous players.
- All decisions shall be generated programmatically according to the assigned player behaviours.
- Random events shall be generated using an appropriate pseudo-random number generator.
- All monetary calculations shall be performed using integer values.
- Every board space shall be represented using appropriate data structures.
- Every player shall maintain independent financial records.
- Every property shall maintain ownership, development, insurance, mortgage, depreciation, and valuation information.
- The simulation shall terminate when only one solvent player remains or when 500 rounds have elapsed.
- The winner shall be determined according to the rules specified earlier.

The implementation shall consist of at least the source files specified in Table 5.

Table 5: Source Files Requirement

File	Purpose
types.h	Definitions of structures, enumerations, constants, and data types
board.c	Board initialization and movement logic
players.c	Player decision-making algorithms
finance.c	Loans, banking, insurance, depreciation, taxation
events.c	Economic events and government regulations
game.c	Game controller and simulation engine
main.c	Program entry point

Students may include additional files if required. Global variables shall be avoided wherever possible. Dynamic memory allocation should be used only where justified. Programs must compile successfully using “`gcc *.c -o monopoly`” without compiler errors.

5 Required Output Messages

The simulation shall generate detailed output describing every significant event as given below.

- **Before the Game Begins**

MONOPOLY-LK Simulation

Player 1 : Aggressive Investor
Player 2 : Conservative Banker
Player 3 : Risk Taker
Player 4 : Opportunistic Trader

Each player begins with LKR 30,000.

- **Determining the First Player**

Aggressive Investor rolls 9.
Conservative Banker rolls 6.
Risk Taker rolls 11.
Opportunistic Trader rolls 5.

Risk Taker will begin the game.

Turn order:

Risk Taker
Opportunistic Trader
Aggressive Investor
Conservative Banker

- **Dice Roll**

Aggressive Investor rolled 8.

- **Player Movement**

Aggressive Investor moves from Square 12 to Square 20.

- **Passing GO**

Aggressive Investor passed GO.
Collected LKR 2,000.
Current Balance : LKR 24,500.

- **Purchasing Property**

Aggressive Investor purchased Galle Fort for LKR 4,500.
Remaining Balance : LKR 18,000.

- **Paying Rent**

Risk Taker landed on Galle Fort.

Rent Paid : LKR 750.

Owner : Aggressive Investor.

- **Building Construction**

Aggressive Investor constructed one house on Galle Fort.

Construction Cost : LKR 1,500.

- **Hotel Construction**

Aggressive Investor upgraded Galle Fort to a Hotel.

- **Obtaining a Loan**

Aggressive Investor obtained a secured loan.

Loan Amount : LKR 15,000.

Collateral :

Galle Fort

Unawatuna

Interest Rate : 8%

Duration : 20 Rounds

- **Loan Repayment**

Aggressive Investor repaid LKR 5,000.

Outstanding Balance :
LKR 11,200.

- **Loan Default**

Aggressive Investor has defaulted.

Collateral has been foreclosed.

Outstanding debt cleared.

- **Purchasing Insurance**

Comprehensive Insurance purchased.

Property : Galle Fort

Premium : LKR 450.

- **Disaster**

Flood occurred.

Affected Property :

Unawatuna.

Insurance Claim Approved.

Compensation Paid :
LKR 2,500.

- **Auction**

Auction Started.

Property :
Nuwara Eliya

Opening Bid :
LKR 3,500.

Risk Taker bids LKR 3,750.

Aggressive Investor bids LKR 4,000.

Conservative Banker withdraws.

Opportunistic Trader withdraws.

Aggressive Investor wins the auction.

- **Economic Event**

Economic Event

Tourism Boom

Southern Province properties increase in value by 15%.

- **Government Regulation**

Government Regulation

Housing Subsidy Introduced.

Construction costs reduced by 30%.

- **Property Depreciation**

Property

Maharagama

has depreciated by 5%.

Current Value

LKR 4,750.

- **Insurance Expiry**

Insurance policy on Galle Fort expires in 3 rounds.

- **Bankruptcy**

Risk Taker has been declared bankrupt.

Remaining assets transferred to the Bank.

- **End of Every Round**

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Round 37 Summary

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Aggressive Investor

Cash : LKR 12,300

Net Worth : LKR 82,500

Properties : 12

Hotels : 3

Outstanding Loan : LKR 6,500

Conservative Banker

Cash : LKR 28,500

Net Worth : LKR 73,800

Properties : 9

Hotels : 1

Outstanding Loan : None

Risk Taker

Cash : LKR 900

Net Worth : LKR 64,000

Properties : 15

Hotels : 4

Outstanding Loan : LKR 18,000

Opportunistic Trader

Cash : LKR 20,100

Net Worth : LKR 77,200

Properties : 10

Hotels : 2

Outstanding Loan : LKR 3,500

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- End of Game

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GAME OVER

Winner

Aggressive Investor

Total Cash

LKR 65,400

Total Property Value

LKR 210,700

Outstanding Loans

None

Net Worth

LKR 276,100

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• Rule-LK 36 output messages

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Current Market Conditions

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Market Boom

Southern Province (+20%)
Rounds Remaining : 7

Market Decline

Western Residential (-15%)
Rounds Remaining : 4

Regional Development

Northern Development Programme
(+30%)
Rounds Remaining : 12

Inflation

+5%

Current Loan Interest

9%

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UCSC

A National Event Cards

The simulation shall contain a deck of twenty (20) National Event Cards. Whenever a player lands on an Event square, the top card shall be drawn. After execution, the card shall be returned to the bottom of the deck. These effects are applicable to the player who drew the card for 15 rounds in addition to other modifiers.

Card	Effect
Tourism Hype	Hotels earn double rent for 5 rounds
Fuel Shortage	Railway rent doubles for 5 rounds
Heavy Floods	Random coastal property damaged
Political Rally	One random property closed for 2 rounds
Stock Market Rise	All property values increase by 10%
Economic Downturn	Property values decrease by 15%
Housing Subsidy	House construction cost reduced by 30%
Interest Rate Cut	Loan interest reduced by 2%
Interest Rate Increase	Loan interest increased by 2%
Tax Amnesty	Each player receives LKR 2,000
Power Failure	Utility income halved for 3 rounds
Foreign Funding	Commercial property values increase by 15%
Port Expansion	Railway station values increase by 20%
Festival Season	Hotels receive 50% additional rent
Labour Strike	Construction suspended for 2 rounds
Insurance Discount	Premiums reduced by 20%
Property Revaluation	Random property group appreciates by 15%
Currency Depreciation	Construction costs increase by 10%
Government Grant	Random player receives LKR 5,000
National Disaster	Random developed property damaged

B Property Values

Group	Purchase Price	House Cost	Hotel Cost	Mortgage Value
Brown	1,500	500	2,000	750
Light Blue	2,500	750	3,000	1,250
Pink	3,500	1,000	4,000	1,750
Orange	4,500	1,250	5,000	2,250
Red	5,500	1,500	6,000	2,750
Yellow	6,500	2,000	8,000	3,250
Green	8,000	2,500	10,000	4,000
Dark Blue	10,000	3,000	12,000	5,000

C Rent Tables

Table 6: Residential Properties

Development	Rent Multiplier
No Buildings	$1 \times$ Base Rent
1 House	$2 \times$
2 Houses	$3 \times$
3 Houses	$5 \times$
4 Houses	$7 \times$
Hotel	$10 \times$

Table 7: Railway Stations

Stations Owned	Rent (LKR)
1	250
2	500
3	1,000
4	2,000

Table 8: Utility Companies

Utilities Owned	Rent
One Utility	$4 \times$ Dice Value
Both Utilities	$10 \times$ Dice Value

D Interest Tables

Loan interest rates vary according to the prevailing economic conditions.

Table 9: Interest Tables

Economic Condition	Annual Interest Rate
Economic Boom	5%
Stable Economy	8%
Moderate Inflation	10%
High Inflation	12%
Economic Recession	15%

E Insurance Policies

Table 10: Insurance Policies

Policy	Premium	Coverage
Basic Property Insurance	5% of Property Value	Fire and Flood (80% repair cost)
Comprehensive Insurance	10% of Property Value	Fire, Flood, Riot, Vandalism, Earthquake (100% repair cost)
Business Interruption Insurance	15% of Property Value	Repair cost and lost hotel rental income for 5 rounds

